

Risk Factor Agent

Implementation Blueprint · n8n Workflow

Version 1.0
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Overview

An n8n-based AI agent that monitors global news daily, reasons over geopolitical and industry events, and suggests risk weight updates for your feature database. You review and approve all changes before any write occurs.

Workflow Diagram



Feedback loop: After each approved run, reasoning and outcomes are written to the [agent_memory](#) table. The next day's run reads this history so the agent avoids duplicate suggestions and calibrates changes over time.

Implementation Plan

Phase	Step	n8n Node / Tool	Output
1 — Read	Fetch all feature weights	Postgres node (SELECT)	JSON: {feature, weight}
1 — Read	Load last 14-day memory	Postgres node/ Vector DB (agent_memory)	Past reasoning context
2 — News	Search news per feature type	HTTP Request → Benzinga API	News summaries + URLs
2 — News	Loop over feature groups	SplitInBatches node	Batched search results

3 — Reason	Build LLM prompt with context	Code node (JS)	Structured prompt string
3 — Reason	Call Claude for weight diff	Claude / OpenAI node	JSON diff array
4 — Approve	Format & send diff to you	Whatsapp\ telegram	With approve links
4 — Approve	Pause until you respond	Wait node + Webhook	Approved/rejected items
5 — Write	Write approved weights to DB	Postgres node (UPDATE)	Updated weight rows
5 — Write	Log to agent_memory table	Postgres node (INSERT)	Audit trail entry

Technical Reference

Memory Layer — agent_memory Table

Stores one row per feature per daily run. The next run reads the last 14 days and injects them into the LLM prompt, preventing duplicate suggestions and enabling trend-aware reasoning.

Column	Type	Purpose
run_date	DATE	Which daily run created this row
feature	TEXT	e.g. country:UAE, industry:Shipping
news_summary	TEXT	Key news that triggered reasoning
reasoning	TEXT	Why agent suggested the change
suggested_weight	INT	Agent's proposed new value
approved	BOOLEAN	Did you approve this change?
approved_weight	INT	Final value written to DB

LLM Reasoning Prompt Structure

Section	Content
System role	You are a risk weight advisor. Scale: 0 = safest, 10 = riskiest. Be conservative — suggest max ± 2 change per cycle unless high-confidence.
Current weights	JSON of all features and their live weight values pulled from DB.
Memory context	Last 14 days of reasoning: what news was seen, what was suggested, what you approved.
Today's news	Tavily summaries per feature group, with source URLs.

Output format	JSON array: { feature, current_weight, proposed_weight, confidence (low/medium/high), reasoning, sources[] }
Guard instruction	Do not re-suggest a change already captured in memory unless a significant new development justifies it.

Key Design Decisions

News Source Benzinga API	Approval Gate Wait + Webhook Native n8n pattern. Webhook fires when you click the approve link in email.	DB Safety Changelog table Every write logs previous value, new value, news source, and run timestamp.
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